Zhiheng Zhang

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Academic Curriculum Vitae | PhD Applications, Fall 2027

Research Interests

Reliable multi-agent software systems; trustworthy verification of agent execution and completion claims; evidence-qualified acceptance under incomplete observation; persistent execution, failure recovery, and reproducible evaluation.

Education

University of Melbourne, Australia

Master of Computer Science | 2025-present; expected completion December 2026

Two-year program: one year of coursework followed by one year of research. Research supervisor: Professor Richard Sinnott. Master's thesis manuscript completed, with evaluation results documented; degree completion remains pending.

University of British Columbia, Okanagan campus, Canada

Bachelor of Science in Computer Science | 2020-2024; degree awarded 2024

Capstone Software Engineering Project (COSC 499): 93/100.

Preprints and Manuscripts

1. Zhiheng Zhang. Omission-Safe Conditional Contracts for Evidence-Closed CI Qualification. 2026. arXiv preprint, arXiv:2610.39241 [cs.SE].
Developed source-derived CI qualification contracts and an independent certificate checker to prevent missing observations from silently weakening evidence requirements. Research based on DS-Hns.

2. Zhiheng Zhang. Architectural Debt Enforcement under Incomplete Observation: A Controlled Study of Ratchets and Evidence Qualification. 2026. arXiv preprint, arXiv:2610.41276 [cs.SE].
Evaluated evidence-qualified architectural debt enforcement on 240 source files and 480 constructed scenarios, examining identity-aware ratchets, observation uncertainty, and provenance binding. Research based on Codex Boss (Boss-A2).

Technical and Research Practice

- Systems architecture: task lifecycles, interface contracts, durable state, failure handling, and multi-device integration.
- Evaluation: reproducible task scenarios, controlled fault injection, trace inspection, recovery/replay analysis, and evidence-backed acceptance.
- Software practice: JavaScript/Node.js, TypeScript, React Native/Expo, Kotlin/Jetpack Compose, Git/GitHub, and AI-assisted implementation with explicit review.
- Research development: problem formulation, experiment design, artifact organization, and technical documentation.

Selected Academic Preparation

Introduction to Machine Learning; Machine Learning Applications for Health; Research Methods; Analysis of Algorithms; Software Engineering.

Research and Systems Experience

Utopia | Independent systems project

- Led architecture, AI-assisted implementation and integration of a multi-agent system combining persistent task execution with evidence-based acceptance.
- Integrated two physical Windows execution workers with Android and web control interfaces, explicit target-device routing, and inspectable cross-host results.
- Maintained failure histories, task-to-evidence links, and replay records to support debugging and reproducible evaluation.
- Completed and froze the integrated successor to DS-Hns and Codex Boss.

Privacy Lens | Master's research

- Developed evidence-bounded Android privacy review under Professor Richard Sinnott's supervision, preserving provenance, missing context, and permitted claims across admission, policy interpretation, storage, and presentation.
- Defined bounded temporal combination and state-continuity conditions for replay, overlap, source isolation, and recovery; distinguished technical observations from unsupported legal conclusions.
- Documented a 495-APK static census and, separately, device workflows covering 233 package identities: 203 completed and 30 retained failures. These support bounded engineering claims, not legal compliance or population-wide accuracy.

DS-Hns | Persistent coding-task execution

- Designed durable checkpoints and resume checks around workspace state, repository identity, and unfinished effects before interrupted tasks continue.
- Used controlled interruption scenarios to distinguish committed-progress recovery from effect/checkpoint gaps that can still cause duplicate actions.
- Completed and froze the runtime; subsequently developed the CI evidence-qualification study listed above.

Codex Boss | Multi-agent orchestration

- Developed multi-agent workflow orchestration around persistent task contracts, bounded recovery, and deterministic acceptance checks.
- Separated agent-reported completion from host-observed execution evidence, making completion claims subject to explicit acceptance checks; subsequently integrated this work into Utopia.
- Completed and froze the platform; subsequently studied architectural debt enforcement and observation uncertainty in Boss-A2.

Proposed Doctoral Focus

I aim to study how interacting AI agents can produce outcomes that remain verifiable when observations are incomplete, execution is interrupted, or deployment conditions change. My proposed focus is on evidence-aware coordination and acceptance: determining when available observations justify a completion claim, when agents should seek additional evidence, and when a system should defer acceptance or recover. I would evaluate these mechanisms through controlled failures, traceable execution records, and reproducible multi-agent tasks. This direction connects to the Cologne group's work on trustworthy AI and interacting agents. I would like to investigate how explicit evidence requirements and uncertainty-aware acceptance can improve the reliability of agent coordination, with stronger guarantees as a research objective.

THE UNIVERSITY OF BRITISH COLUMBIA

TRANSCRIPT OF ACADEMIC RECORD

ENROLMENT SERVICES
2016 - 1874 East Mall
Vancouver, BC Canada V6T 1Z1

Surname: Zhang
Given Names: Zhiheng
Student Number: 54569462
Date of Birth: October 19

Date Printed: August 28, 2024

Issued To:
Zhang, Zhiheng
nicholas_zhang2020@163.com

REGISTRAR

Note: Transcript valid only if bearing the registrar's signature. This transcript is printed on secure paper on a blue background.

UBC Credentials

None to date

Transfer Credits

None to date

Winter Session 2020 - 2021

Bachelor of Science (UBC Okanagan) - **Year 1**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 111	(3.0)	Computer Programming I	73	B	3.0				389	71
1	COSC 122	(3.0)	Computer Fluency	93	A+	3.0				199	82
1	ENGL 150	(3.0)	Introduction to Literary Genre				W	2020/11/11			
1	MATH 100	(3.0)	Differential Calculus with Applications to Physical Sciences and Engineering	94	A+	3.0				357	69
1	PHYS 111	(3.0)	Introductory Physics for the Physical Sciences I	91	A+	3.0				274	80
2	COSC 121	(3.0)	Computer Programming II	64	C+	3.0				264	74
2	COSC 123	(3.0)	Computer Creativity	93	A+	3.0				124	86
2	ENGL 112	(3.0)	Studies in Composition	66	C+	3.0				27	75
2	MATH 101	(3.0)	Integral Calculus with Applications to Physical Sciences and Engineering	65	C+	3.0				341	75
2	MGMT 100	(3.0)	Introduction to Business	66	C+	3.0				136	74

Summer Session 2021

Bachelor of Science (UBC Okanagan) - **Year 1**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	MATH 221	(3.0)	Matrix Algebra	82	A-	3.0				38	68

Winter Session 2021 - 2022

Bachelor of Science (UBC Okanagan) - **Year 2**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 221	(3.0)	Introduction to Discrete Structures	77	B+	3.0				100	73
1	COSC 222	(3.0)	Data Structures	73	B	3.0				194	78
1	GEOG 108	(3.0)	Earth Systems: Weather, Climate, and Life	80	A-	3.0				167	76
1	MATH 200	(3.0)	Calculus III	77	B+	3.0				48	67
2	COSC 211	(3.0)	Machine Architecture	70	B-	3.0				129	67
2	ENGL 203	(3.0)	Topics in Composition	68	B-	3.0				20	85
2	MATH 225	(3.0)	Introduction to Differential Equations	63	C	3.0				48	64
2	STAT 230	(3.0)	Introductory Statistics	72	B	3.0				131	74

Summer Session 2022

Bachelor of Science (UBC Okanagan) - **Year 2**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 322	(3.0)	Introduction to Artificial Intelligence	79	B+	3.0				18	77
1	JPST 100	(3.0)	Beginning Japanese Language I	92	A+	3.0				26	77
2	STAT 124	(3.0)	Business Statistics	87	A	3.0				17	66

Winter Session 2022 - 2023

Surname: Zhang
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Date Printed: August 28, 2024

Winter Session 2022 - 2023 continued...

Bachelor of Science (UBC Okanagan) - **Year 3**

Term	Course		Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC	301	(3.0)	Introduction to Data Analytics	86	A	3.0				144	77
1	COSC	304	(3.0)	Introduction to Databases	45	F	0.0				208	74
1	COSC	310	(3.0)	Software Engineering	86	A	3.0				109	86
1	MATH	302	(3.0)	Introduction to Probability	66	C+	3.0				11	60
2	COSC	303	(3.0)	Numerical Analysis	85	A	3.0				33	81
2	COSC	320	(3.0)	Analysis of Algorithms	82	A-	3.0				145	80
2	COSC	328	(3.0)	Introduction to Networks	62	C	3.0				79	75
2	GEOG	128	(3.0)	Human Geography: Space, Place, and Community	74	B	3.0				216	78
2	JPST	101	(3.0)	Beginning Japanese Language II	82	A-	3.0				21	74

Summer Session 2023

Bachelor of Science (UBC Okanagan) - **Year 3**

Term	Course		Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	INDG	100	(3.0)	Introduction to Decolonization: Indigenous Studies	68	B-	3.0				80	72

Winter Session 2023 - 2024

Bachelor of Science (UBC Okanagan) - **Year 4**

Term	Course		Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC	304	(3.0)	Introduction to Databases	90	A+	3.0				231	79
1	COSC	341	(3.0)	Human Computer Interaction	73	B	3.0				144	83
1	COSC	344	(3.0)	Image Processing and Applications	88	A	3.0				77	82
1	DATA	101	(3.0)	Making Predictions with Data	79	B+	3.0				107	71
2	COSC	414	(3.0)	Computer Graphics	76	B+	3.0				93	73
2	HEAL	100	(3.0)	Introduction and Principles of Health and Wellbeing	79	B+	3.0				294	84
2	MDST	120	(3.0)	Introduction to Computational Art and Design II	84	A-	3.0				26	86
2	PHIL	331	(3.0)	Computer Ethics	84	A-	3.0				79	92

Summer Session 2024

Bachelor of Science (UBC Okanagan) - **Year 4**

Term	Course		Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1-2	COSC	499	(6.0)	Capstone Software Engineering Project	93	A+	6.0				59	86

********** End of Record **********

THE UNIVERSITY OF BRITISH COLUMBIA

The University of British Columbia was established by an Act of the Provincial Legislature in 1908 and opened in 1915. The University is a member of the Universities Canada. UBC Okanagan was established as a campus of The University of British Columbia in April 2005. UBC Okanagan assumed responsibility for several Okanagan University College degree programs and all continuing students in those programs on July 1, 2005. Okanagan University College ceased its educational operations as of August 31, 2005.

EXPLANATION OF STUDENT ACADEMIC RECORD

Academic terms

The University operates on a year system. The academic year is divided into two sessions. Winter Session is divided into two terms, Term 1 from September to December and Term 2 from January to April. Summer Session starts in May and ends in August and is comprised of Term 1 (May and June) and Term 2 (July and August). Courses may span a single term (Term 1 or 2) or a full session (Term 1 and 2) for either session.

Distance Education and Technology (DET) courses begin on one of two starting dates during Winter Session, designated in the Term column as A,B,C or D. During the Summer Session, DET courses have only one starting date and are designated with an A or B in the Term column.

Academic Awards

All academic awards (Fellowships, Scholarships, Prizes and Medals) administered by the University and issued subsequent to April 1975, are noted on the transcript of academic record.

Course Numbering

Courses numbered 100 to 199 are primarily for the first year students. Similarly, courses numbered 200 to 299 are primarily for second year students, courses numbered 300 to 399 are for third year students, and courses numbered 400 to 499 are for fourth year students. Courses numbered 500 to 699 are considered graduate-level. Courses numbered 700 and above are residency courses.

Class Size and Class Average Grades

Effective February 1999, class size and class average grades are shown for most courses. In courses with more than one section, the class size and average are reported for the student's section.

Grading and Course Weight

In May of 1991, The University of British Columbia adopted a percentage grading and credit weighting system. Course weight is expressed in "credits". In general 1 credit represents 1 hour of instruction or 2 to 3 hours of laboratory work per week throughout one term of the Winter Session. A 1-unit course became a 2-credit course. Courses are normally graded on a percentage basis with a corresponding letter grade derived as shown.

Degree Standings and Averages

Standings are expressed as Class 1 (80% or over), Class 2 (65 to 79%), Class P (50 to 64%) or Honours; not used by all faculties. Effective Winter Session 1996/97, the calculation of the standing and average is based on the average percentage grade of all upper-level (300 or higher) courses used to satisfy the degree requirements (excluding courses graded as Pass/Fail); not used by all faculties. This is not applicable to programs in the Faculty of Graduate Studies and Postdoctoral Studies.

Undergraduate Grading Scale

Percent	Letter Grade	Percent	Letter Grade
90–100	A+	64–67*	C+
85–89	A	60–63*	C
80–84	A-	55–59*	C-
76–79	B+	50–54*	D
72–75	B	0–49	F* (Fail)
68–71	B-		

Graduate Grading Scale

Masters Grading Scale		Doctoral Grading Scale	
Percent	Letter Grade	Percent	Letter Grade
90-100	A+	90-100	A+
85–89	A	85–89	A
80–84	A-	80-84	A-
76–79	B+	76-79	B+
72–75	B	72-78	B
68–71	B-	68-71	B-
64–67	C+	0-67	F* (Fail)
60–63	C		
0-59	F* (Fail)		

*Some programs may require highter passing grades; failing grades are assigned a letter grade of "F"

The following standings also apply:

- AEG - aegrotat standing: indicates that the student is granted credit although unable to complete because of illness. A letter grade will also be recorded and that grade converted to a minimum percentage grade for that category for the calculation of averages.
- AUD - audit
- CH - challenge credit
- CIP - course in progress
- CR/D/F - (introduced in 2010/11) excluded from calculation in all averages; granted where a grade is assigned by an instructor but not included on the student's official transcript; grade is converted to CR, D or F standing. CR– credit, satisfactory completion of the requirements of the subject; D– marginal pass (50%–54%), and not available for those programs or courses that define a passing grade as higher than 50%; F – fail
- EX - exempt
- J - see "A" standing as defined in gradeing scale prior to May 1991
- P or F - no grade assigned, excluded from calculation in all averages; P– pass, requirements of subject completed satisfactorily, credit granted where applicable; F – fail
- H - (introduced 1999/00 session) no grade assigned, excluded from calculation in all averages; H – honours, exceeds course require ments, credit granted where applicable
- PLA - credit assigned based on prior learning in subject area.
- S - see "S" standing as defined in grading scale prior to May 1991
- SD - standing deferred; excluded from calculation in all averages
- T - thesis in progress; graduating essay not submitted; course continuing
- W - official withdrawal

PRIOR TO MAY 1991

Course Weight and Hours

Course weight was expressed in "units". In general, 1 unit represented 1 lecture hour per week (or a laboratory period of approximately 2 to 3 hours each week) throughout both terms of the Winter Session. Some faculties did not assign an unit value to some courses. The maximum mark possible (max) indicated the weight of a course. A 3-unit course (3 hours of lectures per week throughout the Winter Session) was graded out of a total of 150 marks. This course approximated a 6 semester-hour or a 9 quarter-hour course.

Grading classifications

- Class 1 - 80% or better (i.e., 120 or more on a maximum of 150)
- Class 2 - 65% – 79% (i.e., 98 to 119 out of 150)
- Class P - 50%– 64% (i.e., 75 to 97 out of 150) except Dentistry, Graduate Studies, Librarianship, Medicine, Nursing and Rehabilitation Medicine where the range is 60% – 64%
- Class F - failure, no credit granted: below 50% except Dentistry, Graduate Studies, Librarianship, Medi cine, Nursing and Rehabilitation Medicine: below 60%

The following standings also apply:

- A - adjudicated pass: indicates that credit is granted and the course need not be repeated although it may not normally qualify as a prerequisite for further work. The grade as signed by the instructor is used in calculation of averages
- AEG - aegrotat standing: granted to a student who has been successful in studies but unable to write the final exam because of illness or other extenuating circumstances
- AUD - audit
- C - requirements of subject completed satisfactorily; no grade assigned; unit value granted where applicable; excluded from calculation in all averages
- D - standing deferred; excluded from calculation in all averages
- E - exempt
- N - no credit granted; did not write examination or otherwise complete requirements of the course (discontinued begin ning 1985– 86 session)
- S - supplemental privilege granted; failure; no credit granted; supplementals are not permitted in all faculties
- T - thesis in progress; graduating essay not submitted; course continuing
- W - (introduced in 1988/89 session) official withdrawal
- X - time expired; student did not complete course

Year Standing

In most Faculties each "year" was graded as Class 1, Class 2, Class P, Fail or carried a descriptive clause indicating action taken by the Adjudication Committee.

Degree Standing

Expressed as Class 1, Class 2, Class P, or with Honours; not used by all faculties. In some faculties the calculation of standing was based on the studies of the final two years, in others on the final year only.

THIS DOCUMENT CONTAINS SECURITY FEATURES. HOLD AT AN ANGLE TO VERIFY A WATERMARK BASKET WEAVE DESIGN

Undergraduate graduation certificate

Bachelor of Science | University of British Columbia



The University of British Columbia

Kelowna, Canada

The Chancellor of The University of British Columbia
with the approval of the Okanagan Senate upon the recommendation of

The Faculty of Science

has conferred the degree of

Bachelor of Science

Major in Computer Science

on

Zhiheng Zhang

In Witness Whereof and by the authority duly committed
to us we have hereunto set our hand and seal
August 2024

The University of British Columbia

Chancellor

President

Registrar

Dean

Master's academic record

UNOFFICIAL ACADEMIC RECORD

Applicant-prepared summary of available results. An updated official transcript is not currently available; this summary is not an official University of Melbourne transcript.

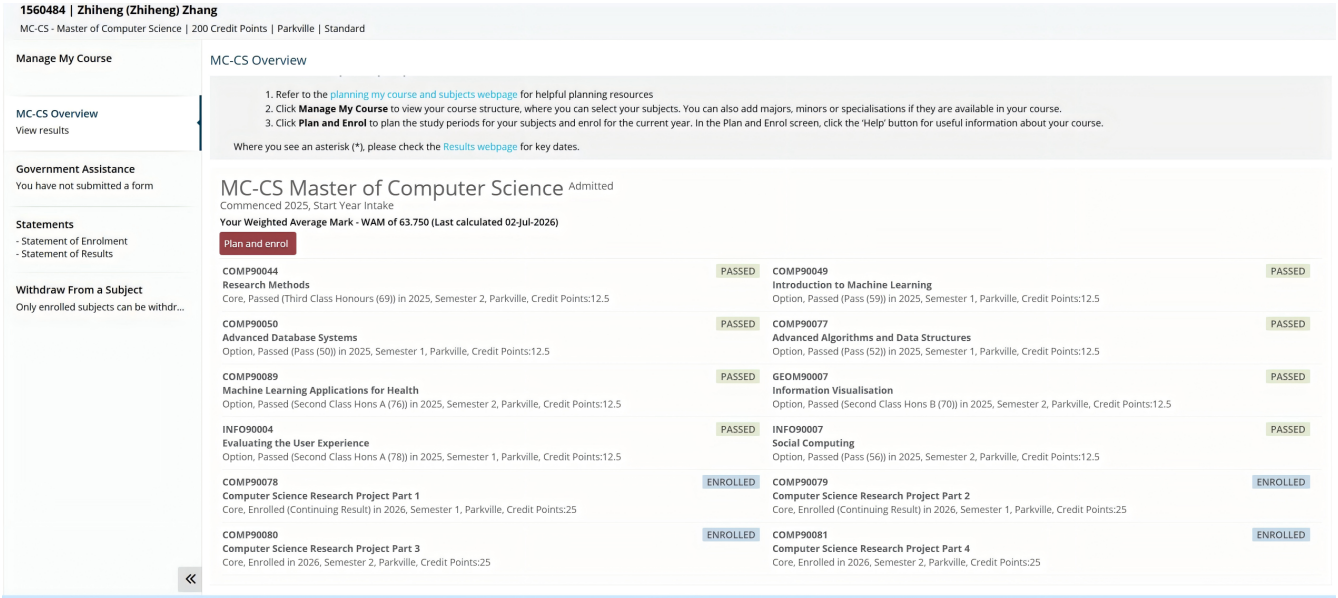
Year / term	Subject	Mark (%)	Credits	Status
2025 / Semester 1	COMP90049	59	12.5	completed
2025 / Semester 1	COMP90050	50	12.5	completed
2025 / Semester 1	COMP90077	52	12.5	completed
2025 / Semester 1	INFO90004	78	12.5	completed
2025 / Semester 2	COMP90044	69	12.5	completed
2025 / Semester 2	COMP90089	76	12.5	completed
2025 / Semester 2	INFO90007	56	12.5	completed
2025 / Semester 2	GEOM90007	70	12.5	completed
2026 / Semester 1	COMP90078	75	25	completed
2026 / Semester 1	COMP90079	75	25	completed
2026 / Semester 2	COMP90080	PENDING	25	pending
2026 / Semester 2	COMP90081	PENDING	25	pending

Derived WAM: 67.5/100 across 150 assessed credit points. This is not a final or officially certified WAM.

COMP90078 and COMP90079 share the combined 30/40 research-project result, normalized to 75% across 50 credits; they are not two independent 30/40 results. COMP90080 and COMP90081 remain pending. The normalization is applicant-calculated and has not been certified by the University.

Master's supporting record

Supporting screenshot supplied by the applicant; not an official transcript.



Year	Study Period	Subject	Course Title	Mark	Credit Point	
2025	Semester 1	COMP 90049	Introduction to Machine Learning	59	12.5	Term Credit = 50,
2025	Semester 1	COMP 90050	Advanced database system	50	12.5	Cumulate Credit = 50
2025	Semester 1	COMP 90077	Advanced algorithm and data structure	52	12.5	Term WAM = (59+50+52+78)/4 = 59.75
2025	Semester 1	INFO 90004	Evaluating User Experience	78	12.5	Cumualte WAM = 59.75 * 50 / 50
						Cumulate WAM = 59.75
2025	Semester 2	COMP 90044	Rasarch Mohods	69	12.5	Term Credit = 50
2025	Semester 2	COMP 90089	Machine Learing Applicatons for Heath	76	12.5	Cumulate Credit = 100
2025	Semester 2	INFO 90007	Social Computing	56	12.5	Term WAM = (69+76+56+70)/4 = 67.75
2025	Semester 2	GEOM 90007	Information Visualization	70	12.5	Cumualte WAM = (59.75 * 50 + 67.75 * 50) / 100
						Cumulate WAM = 63.75
2026	Semester 1	COMP 90078	Computer Science Research Project (Part 1)	30	25	Term Credit = 50
2026	Semester 1	COMP 90079	Computer Science Research Project (Part 2)		25	Cumulate Credit = 150
			This 2 combines, hold 40/100 of Project(1+2+3+4)			Term WAM = 30/40 = 75
						Cumualte WAM = (59.75 * 50 + 67.75 * 50 + 75 * 50) / 100
						Cumulate WAM = 67.5
2026	Semester 2	COMP 90080	Computer Science Research Project (Part 3)	TBD	25	
2026	Semester 2	COMP 90081	Computer Science Research Project (Part 4)		25	
			This 2 combines, hold 40/100 of Project(1+2+3+4)			

Master's supporting record

Supporting screenshot supplied by the applicant; not an official transcript.

Grade	Mark	Description
H1	80%–100%	First Class Honours
H2A	75%–79%	Second Class Honours Division A
H2B	70%–74%	Second Class Honours Division B
H3	65%–69%	Third Class Honours
P	50%–64%	Pass
N	0–49%	Fail
NH	49%	Fail - you hadn't satisfactorily completed all prescribed (hurdle) requirements and would otherwise have passed the subject. A mark of 49% appears on your transcript. For more details, see our hurdle requirements FAQ
S	-	A special or supplementary exam has been awarded, and the exam will be held in the special and supplementary exam period. Special assessment is awarded as an outcome of Special Consideration.
WAF	-	Withheld – assessment to be finalised. This would be used, for example, when: • Special assessment (which will not be a special exam held during the special or supplementary exam period) has been awarded as an outcome of a special consideration application. • Other additional assessment is being considered. • The result has been withheld pending the outcome of an academic misconduct hearing. • Details will be provided via email.
WXT	-	An extension has been granted
CMP	-	Pass (no mark awarded). Only used for subjects marked on a pass/fail basis.

Master's Thesis Abstract

Privacy Lens: Evidence-Bounded Android Privacy Review with Regulation-Driven Temporal Combination Notices

Zhiheng Zhang | University of Melbourne

Supervisor: Professor Richard Sinnott | September 2026

Privacy Lens is a local-first Android research prototype for reviewing privacy evidence without converting technical observations into unsupported legal conclusions. It investigates how provenance, temporal scope, missing context, and permitted claims can remain traceable across evidence admission, policy interpretation, storage, and presentation.

The thesis develops cross-layer evidence contracts and explicit conditions for temporal combination, source and regulation-pack isolation, replay, and recovery. A fail-closed pipeline preserves unavailable evidence rather than treating it as support for a claim. Layered diagnostic evaluation keeps evidence types and denominators separate, retains failures and missing values, and identifies the verification basis of each test suite.

Evaluation includes a hash-pinned census of 495 open-source APKs and, separately, physical-device workflows covering 233 package identities, with 203 completed and 30 retained failures. A single-device interface record provides additional bounded evidence. These results support reproducible engineering claims about evidence handling and traceability, rather than legal compliance or population-wide accuracy. Independent legal review, user comprehension, multi-device generalization, and production readiness remain unestablished. The work offers an evidence-bounded approach to trustworthy verification: making the basis and limits of a system's claims inspectable while preserving uncertainty and failure information.

Preprints and Manuscripts

Zhiheng Zhang

1. Zhiheng Zhang. Omission-Safe Conditional Contracts for Evidence-Closed CI Qualification. 2026. arXiv preprint, arXiv:2610.39241 [cs.SE].
2. Zhiheng Zhang. Architectural Debt Enforcement under Incomplete Observation: A Controlled Study of Ratchets and Evidence Qualification. 2026. arXiv preprint, arXiv:2610.41276 [cs.SE].

Both items are arXiv preprints, not peer-reviewed conference or journal publications.